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High-resolution soybean trial data supporting the expansion of agriculture in Africa

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Understanding how soybean cultivars respond to diverse environmental and management conditions is critical for guiding variety recommendations across African agroecosystems. In this work, we describe a large-scale dataset from multi-environment trials (METs) conducted in Africa over a ten-year period (2015-2024/25). It includes 292 trials carried out in 138 locations across 21 countries, 26,280.00 plots, evaluating 366 soybean varieties. The dataset contains data from morphologic, agronomic and quality traits, as well as detailed environmental information, such as soil features, weather variables, and management practices. We demonstrate the potential of this resource for decision-making in an emerging African soybean production chain by analysing the seed yield across all trials and within specific countries, highlighting cultivar performance and stability. We also applied envirotyping methods to investigate how environmental covariates influence seed yield. This dataset offers a valuable foundation for advancing research on genotype × environment × management (G × E × M) interactions and adaptation of soybean germplasm in Africa, supporting genetic evaluation, environmental modeling, and informed decision-making for soybean expansion across the African continent.

Background & Summary

Soybean (*Glycine max* (L.) Merr.) is a legume crop of major socioeconomic relevance and ranks among the top five food crops globally[1,2]. It provides an affordable source of high-quality plant protein (≈ 40%) and edible oil (≈ 22%), playing an important role in human diets and contributing to the development of profitable agricultural systems[3]. Its by-product, soybean meal, is widely used in key production chains such as poultry, swine, dairy, and aquaculture industries[4]. Globally, Brazil and the United States are the commodity's leading producers. However, in recent years, soybean cultivation has gained increasing attention in Africa as a strategic alternative to generate export revenues that can be reinvested for the benefit of local populations[5]. Advances in breeding programs, genomics, and enviromics approaches have enabled the introduction and expansion of soybeans in several African countries, contributing to the agricultural transformation of the continent and promoting the adaptation of soybeans to diverse African environments[5].

With the global population projected to reach 9.8 billion by 2050[6], there is a growing need for sustainable protein sources to meet rising food demands. Soybeans play a key role in this scenario due to their high protein content and adaptation. The Soybean Innovation Lab (SIL), led by the United States Agency for International Development (USAID), was created to promote soybean development in tropical and subtropical regions, particularly in Africa, by delivering advanced technologies and agronomic information to stakeholders along the value chain. One of SIL's flagship initiatives is the Pan-African Soybean Variety Trials (PATs) (PAT Database), developed in partnership with the African Agricultural Technology Foundation (AATF), the Syngenta Foundation for Sustainable Agriculture (SFSA), and the International Institute of Tropical Agriculture (IITA)[7]. The program aims to identify high-yielding soybean varieties adapted to local conditions and suitable for smallholder farmers. By 2022, SIL had coordinated over 225 multi-location trials across 27 countries in Africa, Central America, Asia, and the Caribbean, evaluating 370 public and private varieties and breeding lines

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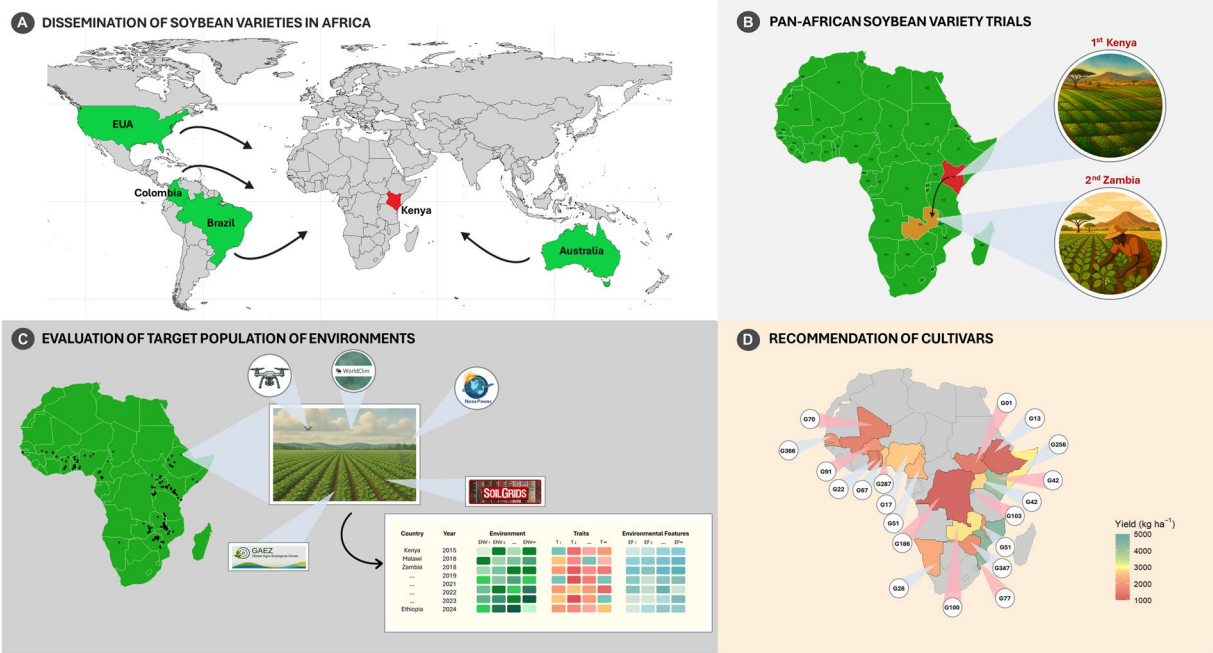

Fig. 1 Illustrative panel depicting the trajectory of the Pan-African Soybean Variety Trials (PAT), coordinated by the Soybean Innovation Lab (SIL). **(A)** The PAT initiative began with the introduction of soybean germplasm into Kenya from countries such as the USA, Colombia, Brazil, and Australia. **(B)** Since 2018, Zambia has served as the primary hub for agronomic evaluation and seed multiplication. After this stage, trials were expanded to multiple African countries to assess varietal adaptation across diverse environmental conditions. **(C)** Evaluation of the target population of environments (TPE), with multi-environment trials conducted between 2015 and the 2024/25 season across 21 African countries, totaling 292 environments and 366 genotypes. Trials followed a randomized complete block design (RCBD) with three replications. Icons represent sources of environmental data used for site characterization (e.g., WorldClim, NASA POWER, GAEZ, SoilGrids, and drone-based sensors). **(D)** Cultivar recommendation based on performance within the TPE. The best-performing genotypes are regionally recommended according to their yield. Genotype numbers and yield scale on the map are illustrative only.

under field conditions. Data collection included agronomic traits and disease incidence under natural infection, contributing to the selection of soybean lines adapted to diverse environments[5].

Initiatives such as the PATs are proofs of how multi-environment trials (METs) are essential tools in modern plant breeding. These trials provide a robust framework for assessing genotype-by-environment (G × E) interaction, which are critical for the effective selection and recommendation of cultivars[8]. In the improvement of quantitative traits, considering G × E allows selection based on phenotypic performance while accounting for both genetic and non-genetic effects[9]. The environments tested in METs are typically sampled from a target population of environments (TPE)—that is, the set of real-world agricultural conditions where new cultivars are expected to be grown[10]. In large-scale networks like PATs, environmental variation spans both space (agroecological zones) and time (growing seasons)[11]. To efficiently exploit G × E in such complex scenarios, one effective strategy is to identify consistent clusters of environments, thereby defining mega-environments. This approach enhances genetic gains by tailoring cultivar recommendations to specific environmental conditions[12].

A holistic understanding of the TPE is essential to support strategic decision-making in plant breeding programs. Measuring the environmental effects that define this population of environments allows the identification and delineation of breeding zones, enabling more efficient allocation of germplasm and precise recommendation of cultivars adapted to specific contexts[13]. Envirotyping tools become fundamental to capture, quantify, and model the environmental variability that influences phenotypic expression and G × E interaction. The integration of these tools with factor-analytic models and management information (such as irrigation regimes or rainfed conditions) enables the delineation of breeding zones based on effective environmental similarity, considering the environment as a dynamic system of interactions among climate, soil, and agronomic practices[14]. This approach allows the optimization of both genotype selection and cultivar recommendation across large cultivation areas, increasing the predictive robustness of models and promoting performance stability across multiple environments[15].

The objective of this work was to explore historical phenotypic data from soybean varieties cultivated across the African continent using a combination of exploratory data analysis, linear mixed models, enviromics, and multivariate approaches. This dataset represents a valuable resource for future research aimed at: (i) identifying stable and high-performing genotypes across diverse agroecological zones; (ii) defining mega-environments based on G × E × M interaction patterns; and (iii) enabling predictive modeling of soybean performance using environmental data and machine learning approaches.

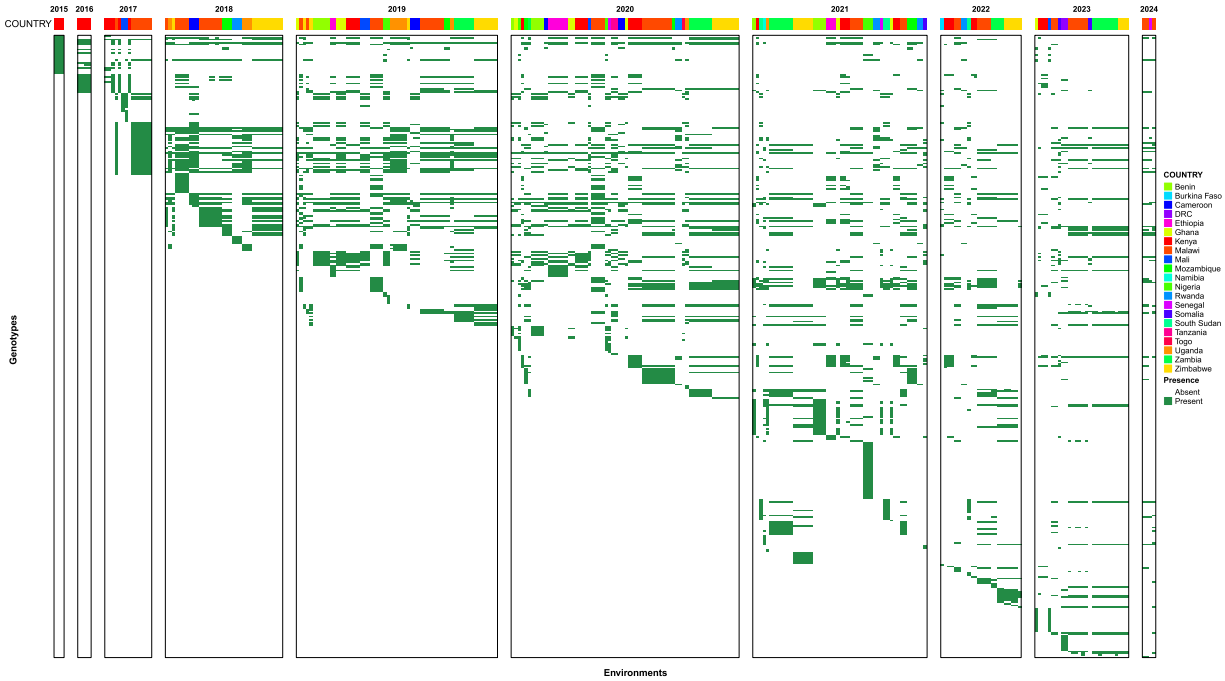

Fig. 2 Genotype-by-environment connectivity heatmap characterizing the multi-environment trials conducted across 21 African countries, encompassing 292 environments and 366 genotypes evaluated over 10 years (2015 to 2024/25). The Y-axis represents genotypes and the X-axis represents environments. The colored bars above the heatmap indicate the countries where trials were conducted. The inner panels are organized chronologically by cropping season. Green cells indicate the presence of a genotype in a given environment.

Methods

Plant Material. The dataset comprises 366 varieties derived from the Pan-African Trials (PATs), coordinated by the Soybean Innovation Lab (SIL). These varieties include both commercial cultivars and experimental lines developed in different countries, selected based on the availability of experimental material from partner institutions, high yield across diverse countries, and desirable agronomic and nutritional qualities[5]. Once the varieties reached Africa, Kenya served as the initial testing site (Fig. 1A), where the varieties were evaluated in field trials[5,16–18]. These early trials helped identify varieties potentially carrying adaptation alleles suited to African agroecological conditions (Fig. 1B). Following their initial evaluation in Kenya, promising varieties were sent to Zambia for seed multiplication. After this multiplication phase, the materials were distributed and tested across other African countries (Fig. 1C), as part of the PATs multi-environment trial network. Following these evaluations, the best-performing and most stable genotypes were selected for further assessment in the subsequent year, with the goal of recommending them to farmers across a wide range of environments and production systems (Fig. 1D).

Experimental Design. Two hundred and ninety-two trials were conducted between 2015 and 2024/25 across 138 locations in 21 countries, encompassing the wide agroecological diversity of the region. Each trial represented a unique combination of site, season, and year. Trials followed a randomized complete block design (RCBD) with three replications. Experimental plots consisted of four five-meter-long rows, spaced 50 cm apart, with approximately 20 plants per row. All trials were manually sown and managed according to local agronomic practices. Irrigation strategies varied across environments: some trials were fully irrigated, others were conducted under rainfed conditions, and a subset followed a supplemental irrigation regime—initially managed as rainfed but receiving additional irrigation during specific critical periods. Phenotypic evaluations were performed on the two central rows, while the outer rows served as borders Fig. 2.

Evaluated Traits. The database comprises three major categories of information: (i) agronomic traits, (ii) nutritional quality, and (iii) a comprehensive set of environmental variables (weather and soil class).

Agronomy. The agronomic traits evaluated across environments included:

- **Emergence:** Plant emergence was assessed both by absolute count (`PL_EMERG_COUNT`) and percentage (`PL_EMERG_PCT`) in the two middle rows at stage V2.
- **Flowering:** Traits included flowering days after sowing (`FLW_DAYS`), and flower color (`FLW_CL`).
- **Maturity:** Maturity plant height (`PH_R8`), and pubescence color (`PUB_CL`) were recorded at stage R8.
- **Lodging and Shattering:** Plant lodging (`LOD`) was scored on a 1–5 scale, where 1 indicates all plants are erect and 5 indicates all plants are completely lodged. The intermediate scores represent increasing levels of

lodging: 1 = all plants erect; 2 = less than 25% lodged (slight inclination); 3 = 25–50% lodged (moderate lodging); 4 = 50–80% lodged (severe lodging); 5 = more than 80% lodged (complete lodging). Pod shattering (`Pod_Shattering`) was evaluated two weeks post-maturity.

- **Yield Components:** Grain yield (`GY`) was measured at 13% moisture, and 100-seed weight (`W100G`) was obtained after seed drying.
- **Disease Resistance:** Foliar disease severity was scored at the R6 stage for multiple pathogens, including bacterial blight (*Pseudomonas savastanoi* pv. *glycinea*) (`R6_BB_Sev`), pustule (*Xanthomonas axonopodis* pv. *glycines*) (`R6_BP_Sev`), tan spot (*Pyrenophora tritici-repentis*) (`R6_BTS_Sev`), cercospora leaf blight (*Cercospora kikuchii*) (`R6_CLB_Sev`), downy mildew (`R6_DM_Sev`), mustard leaf spot (`R6_MLS_Sev`), frogeye leaf spot (*Cercospora sojina*)(`R6_FELS_Sev`), red leaf blotch (`R6_RLB_Sev`), rust severity (*Phakopsora pachyrhizi*) (`R6_RUST_Sev`) and rust reaction type (`R6_RUST_RRT`).

These variables were chosen to comprehensively evaluate genotype performance under natural conditions of biotic and abiotic stress, enabling selection for agronomic stability, resistance, and productivity. For further details, see Supplementary Table 1.

Nutritional Quality - Oil and Protein Content. Seed composition was analyzed using Near-Infrared (NIR) Spectroscopy, following standard protocols from the PAT and IITA soybean breeding programs. Whole-seed samples (50 g per plot) were scanned using a Perten DA 7250 analyzer (PerkinElmer, Inc.), which provides high-throughput predictions based on annual calibration models developed by the University of Minnesota through the Soybean NIR Consortium. Each calibration model includes over 6,000 reference samples with known protein and oil concentrations[17]. Moisture content was measured prior to analysis and reported on an as-is basis. Protein and oil concentrations were expressed on a dry matter basis and converted to grams per kilogram for statistical analysis.

Weather Data. We collected daily-resolution data for nine climatic variables from the NASA POWER (Prediction Of Worldwide Energy Resources) database[19], covering the crop growth period of each trial. The variables included: surface longwave radiation (ALLSKY_SFC_LW_DWN), surface shortwave radiation (ALLSKY_SFC_SW_DWN), corrected precipitation (PRECTOTCORR), relative humidity at 2 meters (RH2M), mean air temperature at 2 meters (T2M), dew point temperature at 2 meters (T2MDEW), daily maximum temperature (T2M_MAX), daily minimum temperature (T2M_MIN), and wind speed at 2 meters (WS2M). Data were retrieved based on the geographic coordinates of each trial and the respective sowing and harvest dates (Supplementary Table 1) using the `nasapower` package[20] in R (R Core Team, 2025)[21]. Additionally, nineteen bioclimatic variables and elevation (ALT) were obtained from the WorldClim version 2.1 database at a spatial resolution of 30 arc-seconds[22]. Values were extracted based on the geographic coordinates of the experimental sites using the *raster*[23] and *terra*[24] packages in R.

Soil Data. A total of 45 edaphic variables were obtained from the SoilGrids database (ISRIC)[25], corresponding to the soil depth intervals of 0-5 cm, 5-15 cm, 15-30 cm, and 30-60 cm. These variables include bulk density (bdod), organic carbon (ocd, soc), total nitrogen, pH in water (phh2o), fractions of clay, sand, and silt, coarse fragments (cfvo), as well as soil temperature (soilT) and bioclimatic variables (SBIO) provided by FAO[26]. The unique geographic coordinates of each experimental environment were used to extract values from the georeferenced raster layers. Data extraction was performed in R using the `raster`[23] and `tibble`[27] packages, with the *extract()* function.

Statistical analysis. In the following notation, V denotes the total number of genotypes, indexed by $v = 1, 2, \ldots, V$; J represents the number of environments, indexed by $j = 1, 2, \ldots, J$; and R corresponds to the number of replicates, indexed by $r = 1, 2, \ldots, R$. The total number of phenotypic records is given by $N = \sum_{j=1}^{J} N_j$, where N_j denotes the number of observations collected within environment j. Statistical analyses were conducted using the Restricted Maximum Likelihood (REML) approach[28] to estimate variance components. To predict genetic values across multiple environments, the Best Linear Unbiased Prediction (BLUP) method was applied[29]. Initially, the assumption of normality was assessed through quantile-quantile (Q-Q) plots, following the recommendation of Kozak and Piepho[30]. Residual independence was assumed within trials, while potential heteroscedasticity was addressed by modeling the residual covariance structure appropriately. Seed yield data were used as the primary response variable for G × E interaction modeling and enviromic prediction analyses.

Outlier elimination. Outlier control was performed using the method proposed by Bernal-Vasquez *et al.*[31], which involves the standardization of residuals (studentized residuals) followed by the application of the Bonferroni-Holm test[32] to identify potential outlying observations. Observations flagged as outliers were removed prior to all subsequent analyses described in this study. The file *data.csv* contains the raw, unfiltered dataset, allowing for the exploration and development of alternative strategies for outlier handling. The outlier detection procedure was implemented in the script named *Outlier.R*.

Calculation of the best linear unbiased estimates (BLUEs). The adjusted means of genotypes within environments were obtained by fitting the following single-trial linear model :

$$\mathbf{y} = \mu\mathbf{1} + \mathbf{X}_1\mathbf{b} + \mathbf{X}_2\mathbf{g} + \boldsymbol{\epsilon} \quad (1)$$

Class	Features	ID	Unit	Min	Mean	Max
Geographical	Latitude	LAT	degree	−16.25	−13.63	−9.93
	Longitude	LON	degree	32.66	33.91	35.50
	Altitude	ALT	m	71.00	977.10	1415.00
Climatic	BIO1 (Annual Mean Temperature)	wc2.1_30s_bio_1	°C	18.45	21.15	25.50
	BIO2 (Mean Diurnal Range)	wc2.1_30s_bio_2	°C	9.27	11.81	12.69
	BIO3 (Isothermality)	wc2.1_30s_bio_3	—	56.64	58.05	61.78
	BIO4 (Temperature Seasonality)	wc2.1_30s_bio_4	—	160.50	243.20	295.80
	BIO5 (Max Temp of Warmest Month)	wc2.1_30s_bio_5	°C	27.10	30.58	34.90
	BIO6 (Min Temp of Coldest Month)	wc2.1_30s_bio_6	°C	8.20	10.23	16.80
	BIO7 (Temperature Annual Range)	wc2.1_30s_bio_7	°C	15.00	20.36	22.00
	BIO8 (Mean Temp of Wettest Quarter)	wc2.1_30s_bio_8	°C	19.92	22.63	27.78
	BIO9 (Mean Temp of Driest Quarter)	wc2.1_30s_bio_9	°C	16.38	18.32	22.95
	BIO10 (Mean Temp of Warmest Quarter)	wc2.1_30s_bio_10	°C	20.80	23.74	28.38
	BIO11 (Mean Temp of Coldest Quarter)	wc2.1_30s_bio_11	°C	15.25	17.73	22.30
	BIO12 (Annual Precipitation)	wc2.1_30s_bio_12	mm	740.00	918.60	1193.00
	BIO13 (Precip of Wettest Month)	wc2.1_30s_bio_13	mm	154.00	227.80	286.00
	BIO14 (Precip of Driest Month)	wc2.1_30s_bio_14	mm	0.00	2.80	15.00
	BIO15 (Precip Seasonality)	wc2.1_30s_bio_15	—	89.80	114.90	125.10
	BIO16 (Precip of Wettest Quarter)	wc2.1_30s_bio_16	mm	435.00	614.50	695.00
	BIO17 (Precip of Driest Quarter)	wc2.1_30s_bio_17	mm	0.00	12.07	57.00
	BIO18 (Precip of Warmest Quarter)	wc2.1_30s_bio_18	mm	99.00	248.80	363.00
	BIO19 (Precip of Coldest Quarter)	wc2.1_30s_bio_19	mm	2.00	15.93	88.00
	Air temperature at 2 m above ground	T2M	°C	19.28	21.75	26.78
	Maximum air temperature at 2 m above ground	T2M_MAX	°C	27.92	31.10	38.20
	Minimum air temperature at 2 m above ground	T2M_MIN	°C	10.69	13.53	17.38
	Dew point temperature at 2 m above ground	T2MDEW	°C	13.28	14.35	16.41
	Relative humidity at 2 m above ground	RH2M	%	59.26	67.57	75.56
	Wind speed at 2 m above ground	WS2M	m/s	1.56	2.44	3.05
	Corrected Precipitation	PRECTOTCORR	mm	1.99	2.41	3.46
	Surface Longwave Radiation	ALLSKY_SFC_LW_DWN	$MJ/m^2/day$	30.73	31.50	34.06
	Surface Shortwave Radiation	ALLSKY_SFC_SW_DWN	$MJ/m^2/day$	19.98	20.70	21.12
Soil	Bulk density 0-5 cm	bdod_0-5cm_mean_30s	g/cm^3	1.30	1.41	1.50
	Bulk density 5-15 cm	bdod_5-15cm_mean_30s	g/cm^3	1.30	1.42	1.50
	Bulk density 15-30 cm	bdod_15-30cm_mean_30s	g/cm^3	1.30	1.43	1.50
	Bulk density 30-60 cm	bdod_30-60cm_mean_30s	g/cm^3	1.30	1.44	1.50
	Coarse fragments 0-5 cm	cfvo_0-5cm_mean_30s	%	1.00	1.90	6.00
	Coarse fragments 5-15 cm	cfvo_5-15cm_mean_30s	%	1.00	1.93	6.00
	Coarse fragments 15-30 cm	cfvo_15-30cm_mean_30s	%	1.00	3.27	8.00
	Coarse fragments 30-60 cm	cfvo_30-60cm_mean_30s	%	2.00	4.87	12.00
	Clay 5-15 cm	clay_5-15cm_mean_30s	%	12.00	18.40	33.00
	Clay 15-30 cm	clay_15-30cm_mean_30s	%	16.00	21.93	41.00
	Clay 30-60 cm	clay_30-60cm_mean_30s	%	19.00	25.70	49.00
	Nitrogen 0-5 cm	nitrogen_0-5cm_mean_30s	g/kg	1.10	1.32	1.60
	Nitrogen 5-15 cm	nitrogen_5-15cm_mean_30s	g/kg	0.90	1.13	1.60
	Nitrogen 15-30 cm	nitrogen_15-30cm_mean_30s	g/kg	0.70	0.89	1.20
	Nitrogen 30-60 cm	nitrogen_30-60cm_mean_30s	g/kg	0.40	0.64	0.90
	Organic carbon density 0-5 cm	ocd_0-5cm_mean_30s	kg/m^3	1.70	2.12	2.60
	Organic carbon density 5-15 cm	ocd_5-15cm_mean_30s	kg/m^3	1.40	1.76	2.60
	Organic carbon density 15-30 cm	ocd_15-30cm_mean_30s	kg/m^3	1.00	1.35	1.80
	Organic carbon density 30-60 cm	ocd_30-60cm_mean_30s	kg/m^3	0.60	0.80	1.20
	Soil pH 0-5 cm	phh2o_0-5cm_mean_30s	—	5.40	6.13	6.40
	Soil pH 5-15 cm	phh2o_5-15cm_mean_30s	—	5.40	6.11	6.40
	Soil pH 15-30 cm	phh2o_15-30cm_mean_30s	—	5.30	6.11	6.40
	Soil pH 30-60 cm	phh2o_30-60cm_mean_30s	—	5.30	6.21	6.90
	Sand 0-5 cm	sand_0-5cm_mean_30s	%	52.00	71.73	80.00
	Sand 5-15 cm	sand_5-15cm_mean_30s	%	51.00	71.67	81.00
	Sand 15-30 cm	sand_15-30cm_mean_30s	%	45.00	68.53	77.00
	Sand 30-60 cm	sand_30-60cm_mean_30s	%	39.00	65.00	74.00
	Silt 0-5 cm	silt_0-5cm_mean_30s	%	7.00	9.97	18.00
	Silt 5-15 cm	silt_5-15cm_mean_30s	%	7.00	9.87	17.00
	Silt 15-30 cm	silt_15-30cm_mean_30s	%	6.00	9.47	18.00

Continued

Class	Features	ID	Unit	Min	Mean	Max
	Silt 30-60 cm	silt_30-60cm_mean_30s	%	6.00	9.13	17.00
	Soil organic carbon 0-5 cm	soc_0-5cm_mean_30s	g/kg	12.00	16.43	20.40
	Soil organic carbon 5-15 cm	soc_5-15cm_mean_30s	g/kg	9.30	14.11	20.00
	Soil organic carbon 15-30 cm	soc_15-30cm_mean_30s	g/kg	6.80	9.34	12.90
	Soil organic carbon 30-60 cm	soc_30-60cm_mean_30s	g/kg	4.80	6.78	8.90
	Soil water content 10 kPa	wv0010	—	249.00	308.30	383.00
	Soil water content 33 kPa	wv0033	—	184.00	230.90	331.00
	Soil water content 1500 kPa	wv1500	—	68.00	104.20	199.00
	Soil temperature (avg)	tsoil	°C	226.17	253.46	292.83
	Temperature seasonality	sts	°C	86.10	155.20	255.70
	Isothermality	iso	—	−84.60	13.67	30.70
	Mean diurnal range	mdr	—	−2.00	1.17	2.40
	SBIO2 Mean Diurnal Range (5-15 cm)	SBIO2_Mean_Diurnal_Range_5_15cm	—	1.50	2.29	3.10
	SBIO3 Isothermality (5-15 cm)	SBIO3_Isothermality_5_15cm	—	11.40	15.31	19.10
	SBIO4 Temperature Seasonality (5-15 cm)	SBIO4_Temperature_Seasonality_5_15cm	—	349.90	371.10	449.80
	SoilT_1 (5-15 cm)	soilT_1_5_15cm	°C	193.00	228.10	289.00
	SoilT_2 (5-15 cm)	soilT_2_5_15cm	°C	188.00	222.20	283.00
	SoilT_3 (5-15 cm)	soilT_3_5_15cm	°C	190.00	229.40	283.00
	SoilT_10 (5-15 cm)	soilT_10_5_15cm	°C	219.00	267.70	313.00
	SoilT_11 (5-15 cm)	soilT_11_5_15cm	°C	235.00	278.60	331.00
	SoilT_12 (5-15 cm)	soilT_12_5_15cm	°C	199.00	242.40	294.00

Table 1. Summary statistics (minimum, mean and maximum) of the 95 environmental covariates recorded for 32 soybean trial environments in Malawi during the 2018-2024/25 cropping seasons. Covariates were classified into *Geographical* (latitude, longitude, altitude), *Climatic* (19 WorldClim v2.1 bioclimatic indices plus NASA-POWER radiation, temperature, humidity, wind-speed and water-balance metrics) and *Soil* (texture, bulk-density, chemical composition, water-holding capacity and soil-temperature layers derived from SoilGrids). Reported values correspond to the range observed across all environments and seasons.

where $\mathbf{y}$ is the vector of phenotypic observations of dimension $N_j \times 1$; μ is the intercept; $\mathbf{1}$ is a vector of ones of length $N_j \times 1$; $\mathbf{b}$ is the vector of design (e.g., block) fixed effects, assumed to follow $\mathbf{b} \sim \mathrm{N}(\mathbf{0}, \sigma_b^2\mathbf{I}_R)$, with incidence matrix $\mathbf{X}_1$ of dimension $N_j \times R$. The vector of genotypic fixed effects is denoted by $\mathbf{g}$, with dimension $V_j \times 1$, and is assumed to follow $\mathbf{g} \sim \mathrm{N}(\mathbf{0}, \sigma_g^2\mathbf{I}_{V_j})$, with incidence matrix $\mathbf{X}_2$ of dimension $N_j \times V_j$. The residual effects are represented by the vector $\boldsymbol{\varepsilon}$, of dimension $N_j \times 1$, and are assumed to follow $\boldsymbol{\epsilon} \sim \mathrm{N}(\mathbf{0}, \sigma_\epsilon^2\mathbf{I}_{N_j})$.

Multi-environment trial analysis. We conducted a MET analysis, following which a linear mixed model was applied:

$$\mathbf{y} = \mu\mathbf{1} + \mathbf{X}_1\mathbf{a} + \mathbf{X}_2\mathbf{b} + \mathbf{Z}_1\mathbf{g} + \boldsymbol{\epsilon} \tag{2}$$

where $\mathbf{y}$ is the vector of phenotypic observations of dimension $N_j \times 1$; μ is the intercept; $\mathbf{1}$ is a vector of ones of length $N_j \times 1$; $\mathbf{a}$ a is a vector of fixed environmental effects, with incidence matrix $\mathbf{X}_1$ of dimension $N \times J$. $\mathbf{b}$ is the vector of block fixed effects, assumed to follow $\mathbf{b} \sim \mathrm{MVN}(\mathbf{0}, \sigma_b^2\mathbf{I}_R)$, with incidence matrix $\mathbf{X}_1$ of dimension $N_j \times R$. $\mathbf{g}$ is a vector of random genotypic effects with dimension $V_j \times 1$, and is assumed to follow $\mathbf{g} \sim \mathrm{MVN}(\mathbf{0}, \sigma_g^2\mathbf{I}_{V_j})$, with incidence matrix $\mathbf{Z}_1$ of dimension $N_j \times V_j$. $\mathbf{b}$ and $\mathbf{g}$ are nested effects, so their dimensions are $RJ \times 1$ and $VJ \times 1$. The residual effects are represented by the vector $\boldsymbol{\varepsilon}$, of dimension $N_j \times 1$, and are assumed to follow $\boldsymbol{\epsilon} \sim \mathrm{MVN}(\mathbf{0}, \sigma_\epsilon^2\mathbf{I}_{N_j})$.

The genetic covariance matrix $\boldsymbol{\Sigma}_g$ was modeled using a factor analytic (FA) structure[33,34]:

$$\boldsymbol{\Sigma}_g = \boldsymbol{\Lambda}\boldsymbol{\Lambda}^\top + \boldsymbol{\Psi} \tag{3}$$

where $\boldsymbol{\Lambda}^{(J\times K)}$ is the matrix of factor loadings, K is the number of latent factors, and $\boldsymbol{\Psi} = \mathrm{diag}(\psi_1, \ldots, \psi_J)$ is a diagonal matrix of specific variances.

Genotypic effects were modeled as[34,35]:

$$\mathbf{g} = (\boldsymbol{\Lambda} \otimes \mathbf{I}_V)\mathbf{f} + \boldsymbol{\delta} \tag{4}$$

where $\mathbf{f}^{(VK\times 1)}$ is the vector of genotype-specific factor scores (latent variables), and $\boldsymbol{\delta}^{(VJ\times 1)}$ is the vector of lack-of-fit terms or specific deviations[33,34]. Factor rotation was applied as described in Smith *et al.*[36] to improve interpretability. We selected the models using the Akaike Information Criterion (AIC)[37] and the proportion of genetic variance explained by the *kth* factor in environment J (V_{k_J}), as recommended for FA models[38]. These

criteria enabled us to assess both model parsimony and the contribution of latent factors to capturing the G × E interaction structure.

$$\mathrm{AIC} = -2\log L + 2p; \quad \bar{v} = \frac{\mathrm{tr}\,(\mathbf{\Lambda}\mathbf{\Lambda}')}{\mathrm{tr}\,(\mathbf{\Sigma}_g)}; \quad v_{k_j} = \frac{\hat{\lambda}_{k_j}^{\star 2} d_k}{\sum_{k=1}^{K} \hat{\lambda}_{k_j}^{\star 2} d_k + \hat{\psi}_j} \times 100$$

where $\boldsymbol{\lambda}^{\star}$ is the factor loading matrix, d_k is the kth element of the diagonal of $\mathbf{D}$. The environment-specific generalized heritability (H_j^2) was estimated according to Cullis *et al.*[39]:

$$H_j^2 = 1 - \left(\frac{\bar{v}_{\Delta}^{\mathrm{BLUP}}}{2\sigma_{g_j}^2} \right) \tag{5}$$

where $\bar{v}_{\Delta}^{\mathrm{BLUP}}$ is the average pairwise prediction error variance for genotypes in environment J, and $\sigma_{g_J}^2$ is the genetic variance in that environment, obtained from the diagonal of $\mathbf{\Sigma}_g$.

The coefficient of variation (CV_j) was computed as:

$$CV_j = \frac{\sigma_{e_j}}{\mu_j} \tag{6}$$

where σ_{e_j} is the residual standard deviation and μ_j is the mean grain yield for environment j.

The genetic correlation matrix between environments was derived following Cullis *et al.*[40]:

$$\mathbf{\Upsilon} = \mathbf{\Phi}(\mathbf{\Lambda}^{\star}\mathbf{D}\mathbf{\Lambda}^{\star} + \mathbf{\Psi})\mathbf{\Phi} \tag{7}$$

where $\mathbf{\Upsilon}$ is a $J \times J$ matrix of pairwise genotypic correlations; $\mathbf{\Phi}$ is a $J \times J$ diagonal matrix whose elements are the inverses of the square roots of the diagonal elements of $\mathbf{\Lambda}^{\star}\mathbf{D}\mathbf{\Lambda}^{\star} + \mathbf{\Psi}$; $\mathbf{\Lambda}^{\star}$ is the rotated factor loading matrix; $\mathbf{D}$ is the $K \times K$ diagonal matrix of factor score variances ($\mathbf{D} = \{d_k\}$); and $\mathbf{\Psi}$ is the $J \times J$ diagonal matrix of specific variances ($\mathbf{\Psi} = \{\psi_j\}$).

Factor Analytic Selection Tools. Factor analytic mixed models (FA) have become a powerful approach for modeling G × E × M interaction in multi-environment trials (METs). The factor analytic selection tools (FAST) facilitate decision-making using FA models[35,41]. The Overall Performance (OP_v) represents the average predicted performance of a genotype across environments. It is calculated as the average of genotype-specific factor scores projected onto the mean environmental loadings. OP_v is particularly useful for identifying broadly adapted genotypes:

$$OP_v = \frac{1}{J}\sum_{j=1}^{J} \lambda_{1_j}^{\star} f_{1_v}^{\star} \tag{8}$$

The Root Mean Square Deviation (RMSD) measures the stability of a genotype, calculated as the deviation of the predicted performance in each environment from its overall mean. Genotypes with lower RMSD values are considered more stable across environments:

$$RMSD_v = \sqrt{\frac{1}{J}\sum_{j=1}^{J} \epsilon_{vj}^2} \tag{9}$$

where $\boldsymbol{\epsilon}$ is the portion of $\mathbf{g}$ (Eq. (4)) without $\lambda_{1j}^{\star} f_{1v}^{\star}$.

We also calculated the reliability of each genotype following Mrode[42], as:

$$r_v = 1 - \frac{PEV_v}{\overline{\sigma_g^2}} \tag{10}$$

where PEV represents the prediction error variance, derived from the diagonal elements of the inverse of the coefficient matrix in the mixed model equations. We applied a selection index (I_v) based on these metrics[43], defined as:

$$SI_v = \left(2 \times \frac{OP_v - \overline{OP}}{\sqrt{Var(OP)}} - \frac{RMSD_v - \overline{RMSD}}{\sqrt{Var(RMSD)}} \right) \times r_v \tag{11}$$

Envirotyping applied in Soybean Breeding. Altitude and mean temperature data were retrieved for each of the J environments in Malawi to investigate their association with grain yield (Table 1). The eBLUEs were estimated using Equation (1). Additionally, we explored how management practices varied along the environmental gradient. To characterize the environmental context of each trial, we computed the mean value of each

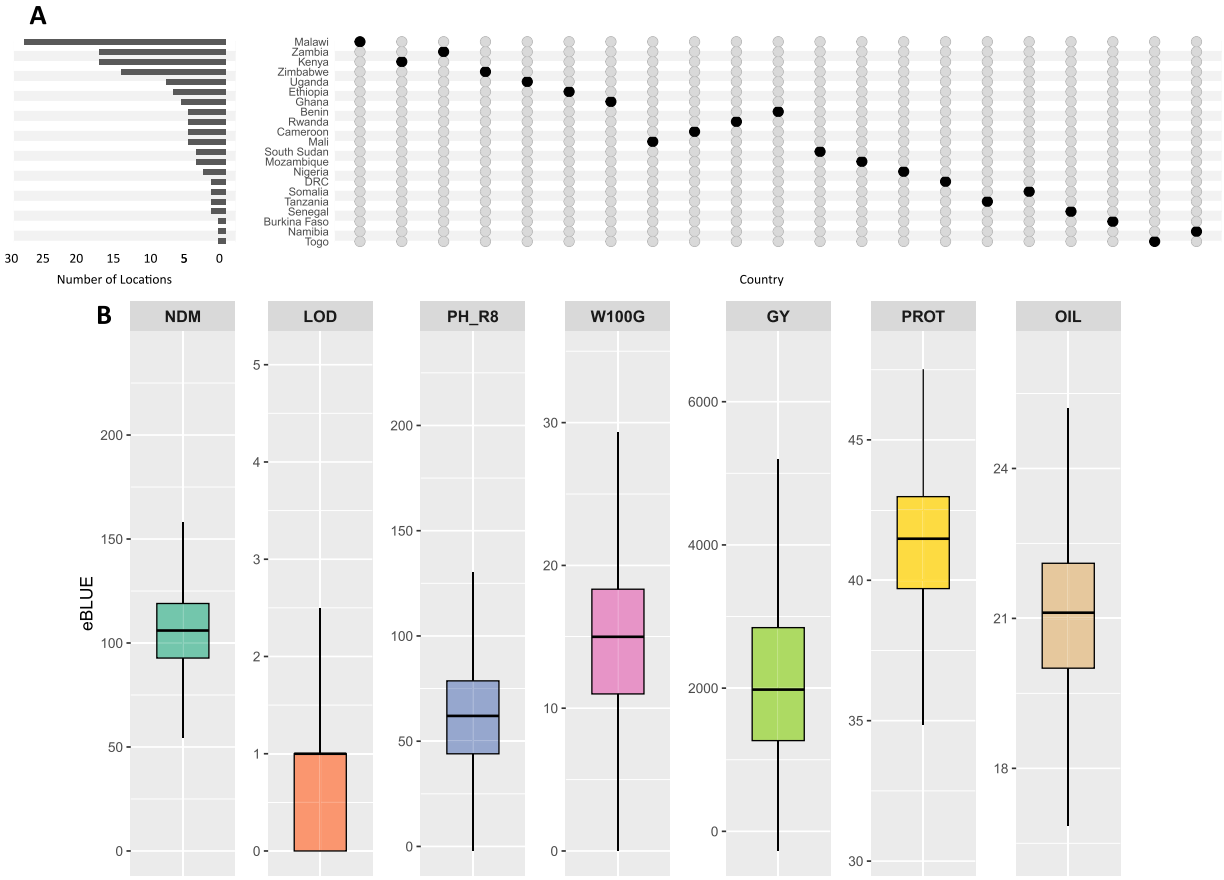

Fig. 3 Characterization of testing intensity and phenotypic variation across 292 multi-environment soybean trials conducted in 21 African countries. (**A**) Number of testing locations per country, summarizing the spatial testing intensity across the Pan-African trial network. (**B**) Distribution of eBLUEs for seven key agronomic traits: number of days to maturity (NDM, days), lodging (LOD, grade scale), plant height at R8 (PH_R8, m), 100-seed weight (W100G, g), grain yield (GY, kg.ha^{-1}), protein content (PROT, %), and oil content (OIL, %).

numeric environmental variable within environments (Table 1). To identify the most relevant environmental drivers of grain yield, we applied sparse Partial Least Squares regression (sPLS) using the `mixOmics`[44] R package. The input response variable consisted of the BLUEs of seed yield across environments. The sPLS model was fitted with two components and a sparsity constraint of 10 variables retained per component. This approach jointly performed dimensionality reduction and variable selection. Variable importance was assessed using the Variable Importance in Projection (VIP) scores. The mean VIP across the two components was used to rank environmental features[44]. The ten most influential variables were visualized by integrating component loadings and VIP values in a bivariate bar plot, where higher VIPs indicate greater relevance in predicting seed yield.

Data Records

The dataset used in this study is available at the Figshare[45] repository (https://doi.org/10.6084/m9.figshare.30257359). The repository is organized into six main directories, each serving a distinct purpose within the workflow. The `data` directory contains the key datasets employed in the analyses, including `data.csv`, which comprises raw genotypic data from the evaluated soybean trials, and `Covamb.csv`, which compiles environmental covariates characterizing 292 multi-environment trials. Additional files include variables related to elevation, weather, soil properties, bioclimatic indices, and geographic coordinates of the trials. The `docs` directory provides supplementary documentation with detailed descriptions of the trials, experimental designs, sowing dates, and local protocols. The `figures` folder includes all graphical outputs generated throughout the analysis, such as correlation plots, boxplots, factor analytic (FA) biplots, partial least squares (PLS) results, and thematic maps, all available in PDF, JPG, and PNG formats.

The `metadata` directory is subdivided into two parts: the `articles` subfolder contains peer-reviewed publications associated with the dataset, while the `docs` subfolder includes a file named `Supplementary.pdf`, which provides comprehensive descriptions of variables, units of measurement, and methodological details. Referenced studies include Meyer *et al.*[17], Marcillo *et al.*[18], Favoretto *et al.*[16], Stella *et al.*[46] and Araújo *et al.*[5]. The `scripts` directory hosts all R scripts used in the analytical pipeline, including those for adjusted means estimation (`Residual.R`), outlier detection and filtering (`Outliers.R`), factor analytic modeling (`Fa_Model.R` and `fa4.R`), environmental data processing (`weather.R`, `SoilData.R`, and `altitude.R`), and PLS modeling

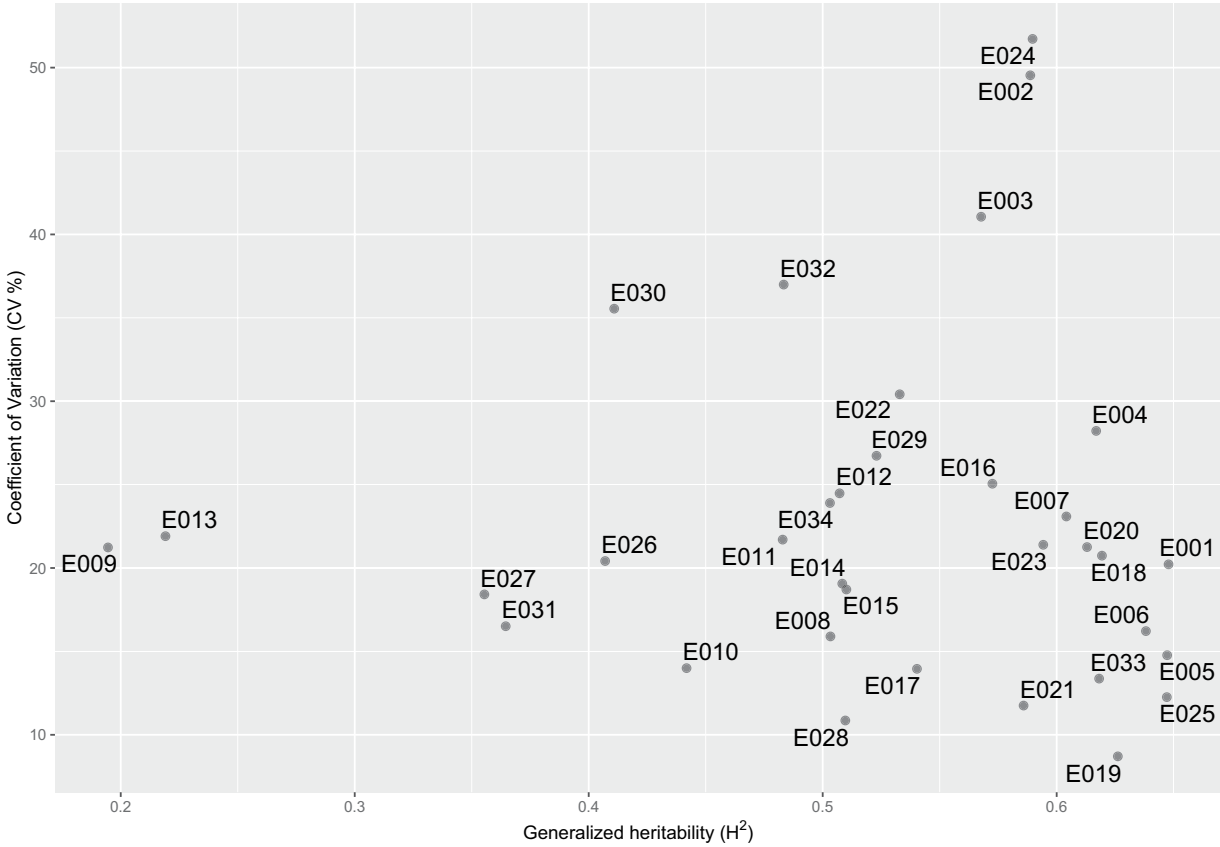

Fig. 4 Relationship between generalized heritability (X-axis) and the coefficient of variation (CV, Y-axis) for environments conducted in Malawi from the 2015 to 2024/25 growing seasons.

with visualization (`PLS.R`, `Map.R`, and `boxplot.R`). Finally, the `output` directory stores model outputs and intermediate results, such as `Models.RData` containing fitted model objects, `eblues_all.RData` with empirical best linear unbiased estimates (eBLUEs) for genotypes, and files like `fa4.RData` and `fa_res.RData`, which correspond to results from the factor analytic models.

Technical Validation

Technical Validation I - Overview of the complete dataset. We analyzed data on the seven most relevant agronomic and nutritional quality traits of soybean (section), collected from trials conducted across 21 African countries between the 2015 and 2024/25 growing seasons. Malawi, Zambia, Kenya, and Zimbabwe received the majority (62.33%) of the trials (Fig. 3A). In contrast, protein (PROT) and oil (OIL) content exhibited narrower distributions, indicating lower phenotypic variability. Lodging (LOD) highlighted the presence of genotypes with increased susceptibility to lodging, which is a critical consideration for breeding programs. This dataset provides a robust foundation for developing balanced selection strategies aimed at identifying superior genotypes that integrate both desirable agronomic performance and nutritional quality.

Technical Validation II - Genotype-by-environment interaction in Malawi. For Technical Validation II, we used grain yield data from trials conducted in Malawi across the 2015 to 2024/25 growing seasons. After applying the likelihood ratio test (LRT). We analyzed a total of 34 environments, defined as the combination of season, location, and year. We found no strong correlation between H^2 and CV%, suggesting that trials with higher experimental variability may still exhibit good precision. Generalized heritability ranged from 30% to 50% in most environments, indicating a moderate capacity to discriminate among genotypes. Additionally, most trials exhibited CV values below 30%, indicating acceptable experimental quality for genotypic inference Fig. 4.

The genetic correlation between environments revealed distinct patterns of G × E interaction. We observed high positive genetic correlations ($r > 0.90$) between several pairs of environments, indicating highly similar genotypic responses. For instance, Bwanje-2020 (E001) and Domisi-2020 (E002) showed a correlation of approximately 0.99, while Bwanje-2020 and Chilanga-2020 (E005) reached 0.97. Other highly correlated pairs included Chilanga-2020 with Bvumbwe-2020 (E006), Chilanga-2020 with E004, and Chilanga-2020 with E003, all showing correlations above 0.90 (Fig. 5). These strong positive correlations suggest that genotype performance was stable across these 2020 environments. In contrast, strongly negative genetic correlations ($r < -0.70$) were observed between environments from different years and regions, reflecting crossover interaction. This pattern was especially prominent between Bwanje-2020 (E001) and several 2021 environments, including Domasi-2021 (E014) with $r = -0.95$, Kasungu-2021 (E017) with $r = -0.88$, Dowa-2021 (E016) with $r = -0.77$,

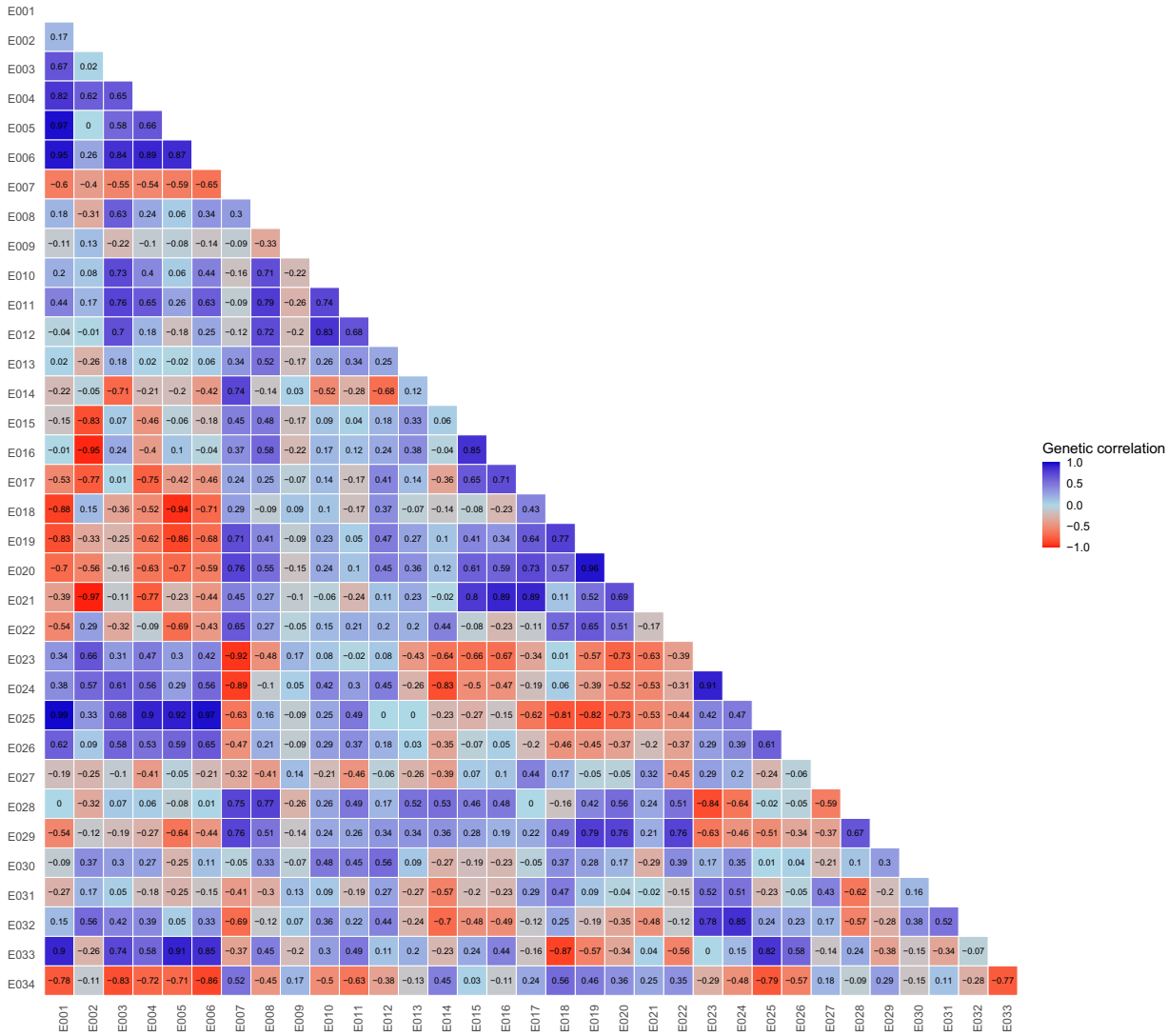

Fig. 5 Heatmap of genetic correlation between pairs of environments, ranging from -1 (red) to +1 (dark blue).

and Dedza-2021 (E013) with $r = -0.83$. These results highlight contrasting environmental conditions between years and regions, where genotypes that performed well in one location-year combination tended to perform poorly in others. Such patterns reinforce the importance of implementing region- and season-specific selection strategies to optimize cultivar recommendation.

The scatterplot of overall performance (OP) versus root mean square deviation (RMSD) provides a comprehensive assessment of genotypic performance and stability across the evaluated environments. G072 was the top performer, combining the highest mean performance with intermediate RMSD, suggesting strong potential for broad adaptation despite moderate variability. Similarly, G002, G025, and G130 showed favorable performance and moderate stability, indicating their suitability as promising candidates for selection. In contrast, G063 exhibited a very low RMSD, reflecting high stability, although its OP was moderate. Such profiles are desirable in environments where stability is prioritized over yield. Mean while, G012, G093, G039, and G091 displayed relatively high mean performance but were associated with substantial instability (RMSD > 400), making them more suitable for specific adaptation strategies. The color scale representing prediction reliability revealed that genotypes such as G013, G117, and G052 had high reliability values (>0.75), supporting the accuracy of their genetic estimates and strengthening their candidacy for selection. Genotypes positioned below the horizontal reference line ($Y = 0$) exhibited below-average performance and are not recommended for advancement in subsequent breeding stages. Overall, this analysis enabled the simultaneous evaluation of genotypic performance, stability, and predictive reliability, allowing for the identification of superior and consistent genotypes under diverse environmental conditions Fig. 6.

Technical Validation III - Enviromics Applied in Soybean Breeding. Altitude and mean temperature were included as environmental covariates to investigate their influence on grain yield. The relationship between altitude and mean temperature revealed an expected environmental pattern, with temperature tending to

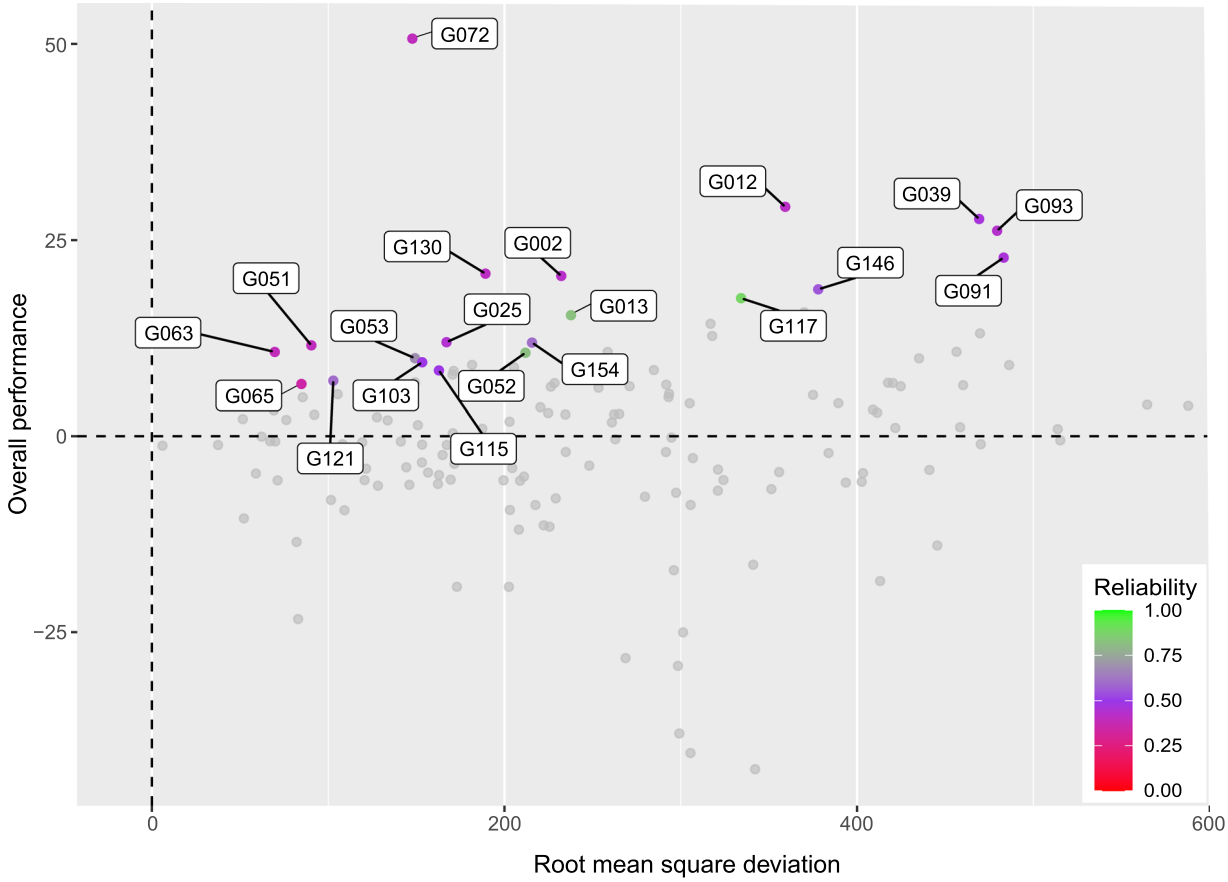

Fig. 6 Plot of Overall Performance (Y-axis) versus Root Mean Square Deviation (RMSD, X-axis) for the genotypes. Genotypes located near the origin of the plot (dashed line) are considered more stable. Genotypes in the upper-left quadrant (high performance and low RMSD) are considered ideal. Colors indicate reliability, ranging from green (high reliability) to purple (low reliability). Labeled genotypes represent of 20% selected.

decrease as altitude increased, reflecting a typical altitudinal gradient. However, grain yield (GY, kg ha^{-1}) exhibited considerable variability along the altitudinal gradient, with no clear linear trend (Fig. 7).

The stratification of data points based on water regime revealed marked differences in yield behavior. Trials conducted under full irrigation generally exhibited higher grain yield and lower variability, even at higher altitudes. In contrast, rainfed environments showed wide dispersion in GY values, with a tendency toward lower mean performance, particularly at intermediate and lower altitudes. Trials under supplementary exhibited intermediate yield values, reinforcing the importance of water availability as a key modulator of crop productivity. These results highlight a complex environmental interaction in which factors such as altitude, mean temperature, and water regime jointly influence genotypic performance. Therefore, interpreting genotypic adaptation in contrasting environments requires a multivariate approach that simultaneously accounts for these variables.

We identified the ten environmental variables most strongly associated with soybean grain yield across trials conducted in Malawi using a sPLS regression model with two components (Fig. 8). The model showed good explanatory capacity, and the variable loadings were used in combination with the variable. The top-ranked variables, based on the average VIP across both components, were predominantly climatic variables derived from atmospheric and biophysical data. Among them, the most important were T2MDEW, BIO2_Mean_Diurnal_Range_5_15cm, SBIO3_Isothermality_5_15cm, ALLSKY_SFC_SW_DWN, and wc2.1_30s_bio_2. In addition, variables related to soil pH at different depths (0-5 cm and 5-15 cm), minimum air temperature (T2M_MIN), and soil moisture (wc2.1_30s_bio_12 and bio_13) also showed VIP scores above 0.7, indicating moderate relevance in explaining grain yield variability. These results suggest that variables related to air and soil temperature, solar radiation, and soil thermal dynamics play a key role in the adaptation and productivity of soybean cultivars across diverse environments in Malawi.

Data availability

All datasets generated and analyzed in this study are openly available at the Figshare repository (https://doi.org/10.6084/m9.figshare.30257359)[45]. The repository contains complete metadata, detailed variable descriptions, and a structured folder organization to facilitate data reuse and reproducibility.

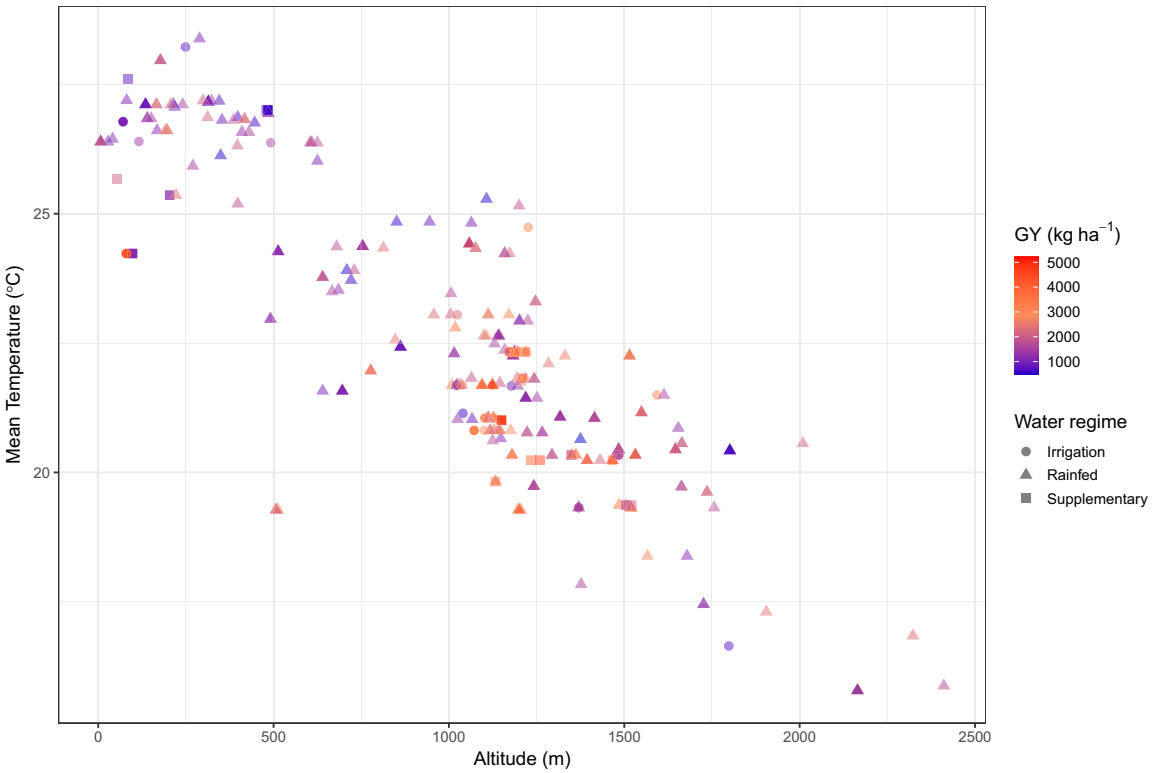

Fig. 7 Relationship between altitude (m) and mean temperature (°C) in multi-environment trials (MET) conducted in Africa. Point color represents grain yield (kg ha^{-1}) during the experimental period, while point shape indicates the water management regime. The color scale ranges from lower to higher yields. The water regimes considered were: irrigated, rainfed, and supplementary (in the latter, the crop was mainly rainfed, with supplemental irrigation when necessary).

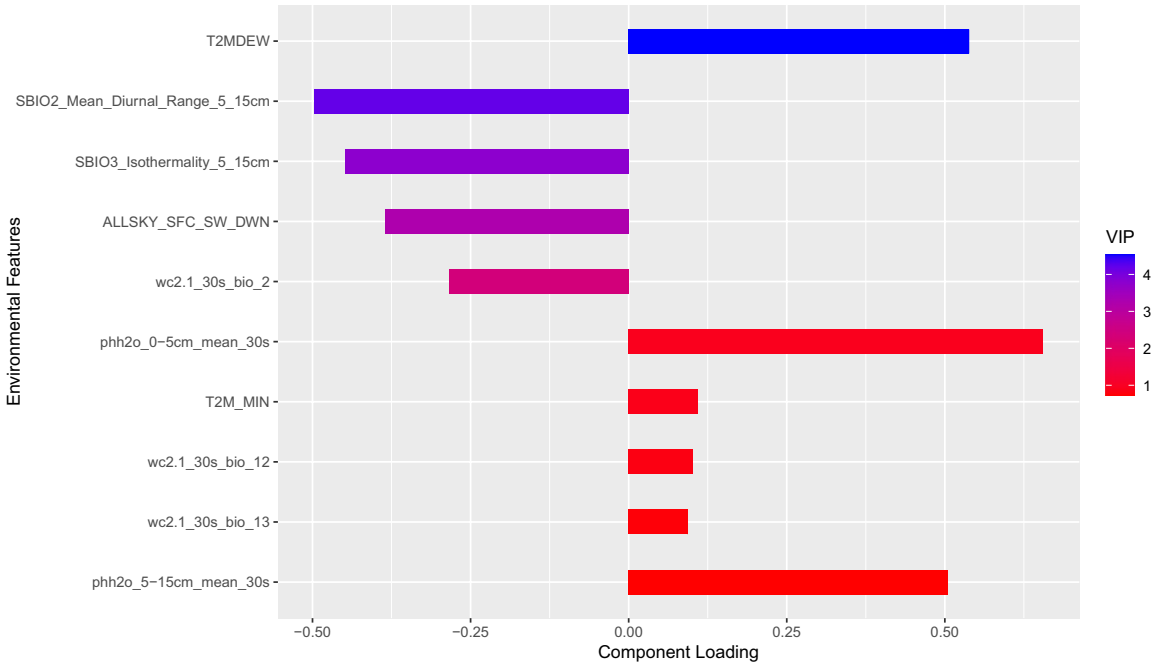

Fig. 8 Top 10 most important environmental features identified by sparse Partial Least Squares (sPLS) regression in relation to grain yield (GY) across soybean trials conducted in Malawi. Variables were selected based on the average Variable Importance in Projection (VIP) across two components. Bar height indicates the loading value of each variable, while the color gradient (red to blue) reflects the mean VIP score, with higher values denoting greater relevance in predicting yield.

Code availability

All data processing and statistical analyses were conducted in the R environment (version 4.5.1). The full set of R scripts, including code for data cleaning, visualization, and technical validation[45], is accessible at https://github.com/mauricioaraujj/Pan_African_Trials_Network.

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Author contributions

M.S.A. led the investigation, conceptualization, supervision, and contributed to writing - review and editing. G.C., B.W.D., T.G., J.N., E.P.L., M.F.S., and P.G. were responsible for conducting the experiments. M.S.A. also performed the data analysis. S.F.S.C., G.N.C.F., and J.B.P. contributed to validation, writing - original draft, supervision, and writing - review and editing. All authors reviewed and approved the final manuscript.

Competing interests

The authors declare no competing interests.

Additional information

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